

Transformations of quadratics



1 Consider the function $y = (x - 6)^2$.

a Complete the table of values.

x	4	5	6	7	8
y					

b Find the minimum value of y .

c Find the x -value that corresponds to the minimum y -value.

d Hence, state the coordinates of the vertex of the parabola.

e How many units to the right has $y = x^2$ been translated?

f Translate the parabola $y = x^2$ to sketch the graph of $y = (x - 6)^2$.

2 Consider the function $y = (x + 6)^2$.

a Complete the table of values.

x	-8	-7	-6	-5	-4
y					

b Find the minimum value of y .

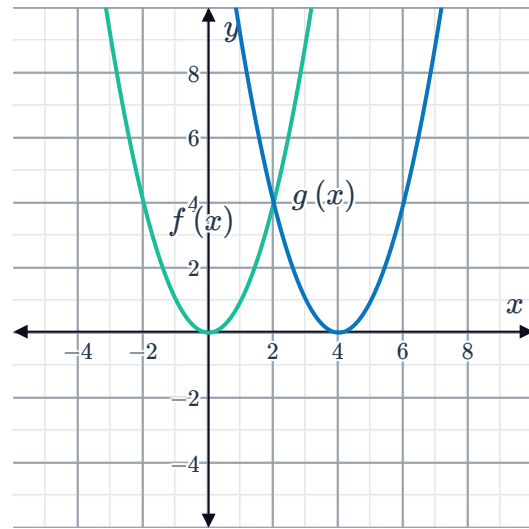
c Find the x -value that corresponds to the minimum y -value.

d Hence, state the coordinates of the vertex of the parabola.

e How many units to the left has $y = x^2$ been translated?

f Translate the parabola $y = x^2$ to sketch the graph of $y = (x + 6)^2$.

- 3 The functions $f(x)$ and $g(x) = f(x + k)$ have been graphed on the coordinate plane.



- a Determine the value of k .
- b Explain what k represents on this context.

- 4 Consider the parabola $y = x^2 - 3$.

- a Complete the table of values.

x	-2	-1	0	1	2
y					

- b Use the graph of $y = x^2$ to sketch the graph of $y = x^2 - 3$.
- c Find the y -intercept of the graph.
- d Adding a constant to the equation $y = x^2$ corresponds to what type of transformation of its graph?

- 5 Consider the following quadratic functions.

- i Sketch the graph of the parabola given by $h(x)$ on a coordinate plane.
- ii Plot the axis of symmetry of the parabola on the same coordinate plane on part (a).
- iii State the vertex of the parabola.

a $h(x) = x^2 - 2$

b $h(x) = x^2 + 2$

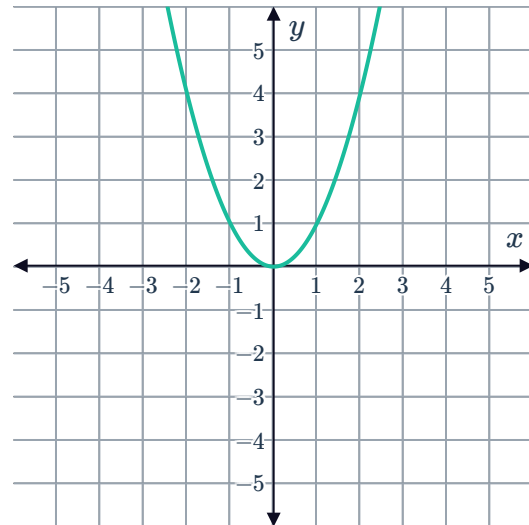
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- a Find the y -intercept of the parabola $y = x^2 + 3$.
- b Find the y -intercept of the parabola $y = x^2 - 4$.
- c How many units must the graph $y = x^2 + 3$ be shifted down to overlap the graph of $y = x^2 - 4$?

7 Consider the given graph of $y = x^2$.



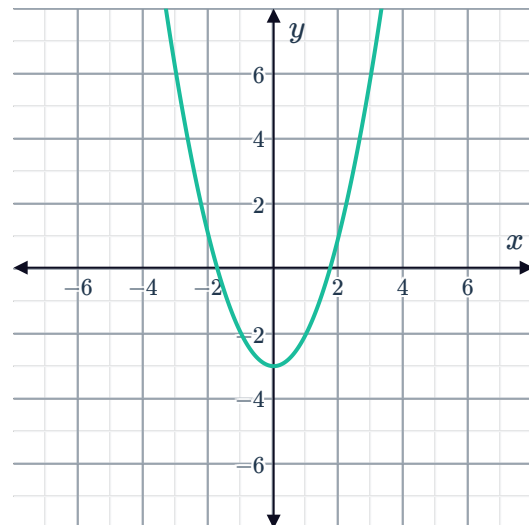
- a
- i How do we shift the graph of $y = x^2$ to get the graph of $y = x^2 - 3$?
 - ii Hence plot $y = x^2 - 3$ on the same graph as $y = x^2$.
- b
- i How do we shift the graph of $y = x^2$ to get the graph of $y = (x - 2)^2$?
 - ii Hence plot $y = (x - 2)^2$ on the same graph as $y = x^2$.



- 8
- a Find the equation of the new parabola when $y = x^2$ is translated 6 units to the right and 2 units down.
 - b Sketch the graph of the parabola formed after the translations.

9 Consider the graph of $y = x^2 - 3$.

- a Determine the new equation when the graph of y is moved upwards by 5 units.
- b Determine the new equation when the graph of y is moved to the right by 3 units.



10 Consider the function $y = -(x - 2)^2$.



a Complete the table of values.

x	0	1	2	3	4
y					

b Find the maximum value of y .

c Find the x -value that corresponds to the maximum y -value.

d Hence, state the vertex of the parabola.

e Translate the parabola $y = x^2$ to sketch $y = -(x - 2)^2$.

11 Consider the parabola $y = -x^2 + 2$.

a Sketch the graph of the parabola.

b State the vertex of the parabola.

12 Consider the two graphs on the same coordinate plane.

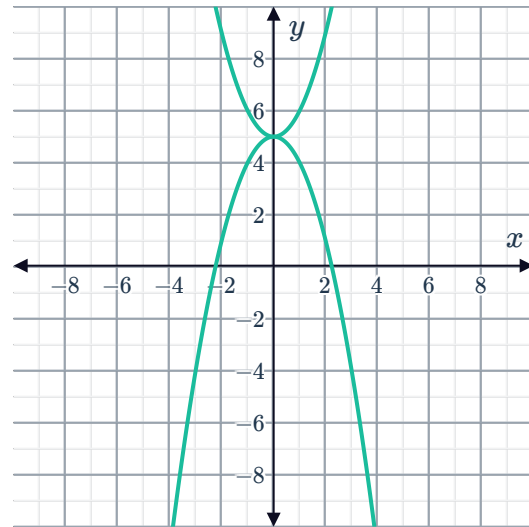
One of them has equation $y = x^2 + 5$.
Determine whether the following could be the equation of the other graph.

a $y = \frac{x^2}{2} + 5$

b $y = x^2 - 5$

c $y = -\frac{x^2}{2} - 5$

d $y = -x^2 + 5$



13 Consider the quadratic function $y = 16 - x^2$.

a Find the x -intercepts.

b Find the y -intercept.

c Sketch the graph of the function.

d Find the vertex of the parabola.

14 Consider the function $y = 2x^2$.



a Complete the table of values.

x	-2	-1	0	1	2
y					

b Sketch the graph on a coordinate plane.

c For $y = ax^2$, as a increases, describe how it changes the graph of $y = x^2$.

15 A parabola of the form $y = ax^2$ goes through $(2, -4)$.

a Determine the value of a .

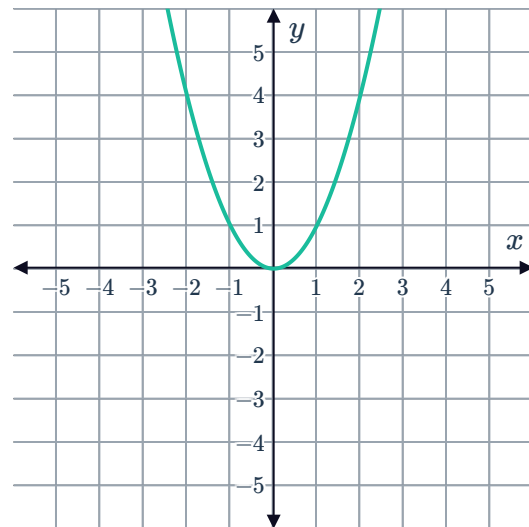
b Find the coordinates of the vertex.

c Sketch the graph of the parabola.

16 Consider the graph of $y = x^2$.

a Sketch the graph of the parabola that has the same vertex as $y = x^2$ and that passes through $(2, 2)$.

b Find the equation of the new parabola.



17 Sketch the graph of the following functions:

a $y = 4x^2 - 2$

b $y = 2(x + 4)^2$

18 Determine whether there has been a reflection on the following transformations.

a $y = x^2$ has been transformed into $y = 5x^2 + 2$.

b $y = x^2$ has been transformed into $y = 3(x - 4)^2$.

c $y = x^2$ has been transformed into $y = 5(x - 1)^2 - 3$.



19 Identify the vertical translation on the following transformations.

- a** $y = x^2$ has been transformed into $y = -2x^2 + 6$
- b** $y = x^2$ has been transformed into $y = -4(x + 10)^2$.
- c** $y = x^2$ has been transformed into $y = 10(x - 9)^2 - 7$.

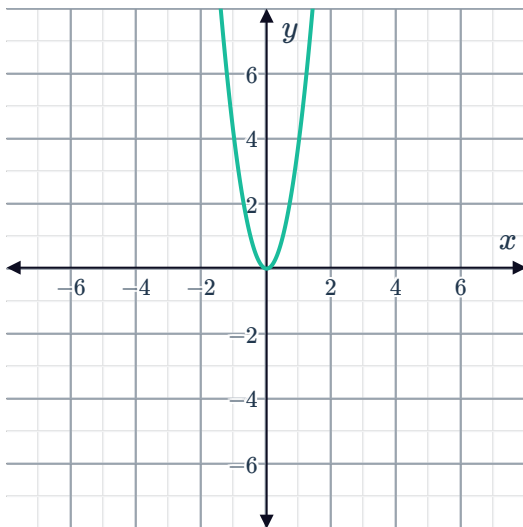
20 Identify the horizontal translation on the following transformations.

- a** $y = -5x^2$ has been transformed into $y = -5(x - 3)^2$.
- b** $y = x^2$ has been transformed into $y = 2(x + 7)^2 + 2$.

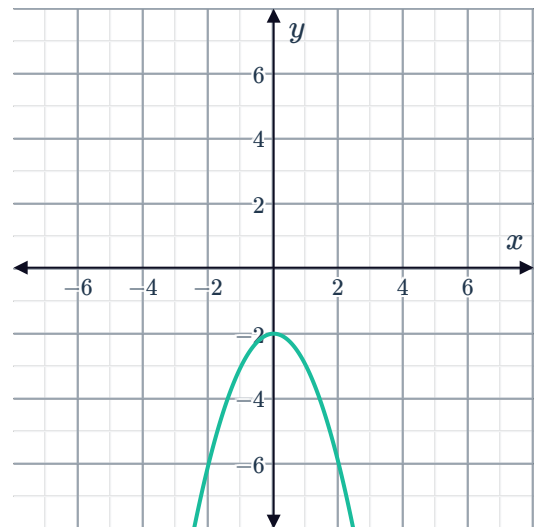
21 For each of the following graphs that has been transformed from $y = x^2$:

- i** Determine the vertical translation.
- ii** Determine the horizontal translation.
- iii** Has there been a reflection about the x -axis?
- iv** Has the quadratic had a change in dilation from the standard $y = x^2$ curve?

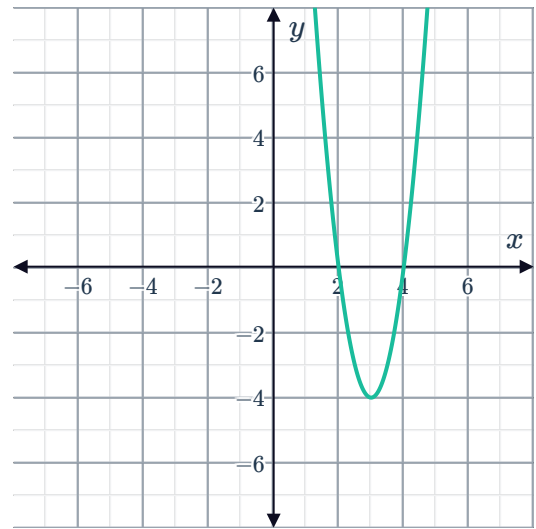
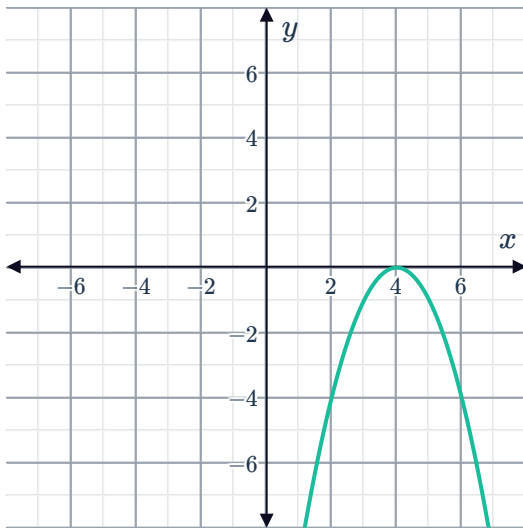
a



b



c



22 Find the resulting equation when $y = x^2$ undergoes the following transformations.

- a** A change in dilation of 10
- b** A change in dilation of 5 and is reflected about the x -axis
- c** A vertical translation of 10 units up
- d** A horizontal translation of 2 units right
- e** A reflection about the x -axis and a vertical translation of 7 units down
- f** A horizontal translation of 4 units left and a change in dilation of 8
- g** Translated horizontally 6 units to the right and translated vertically 5 units upwards

23 Ned graphed the function $f(x) = -4(x - 1)^2 + 2$ and then graphed the function $g(x) = -4(x - 1)^2 - 7$ on the same coordinate plane. Determine whether the following describes the vertex of $g(x)$.

- a** 9 units to the right of the vertex of $f(x)$.
- b** 9 units to the left of the vertex of $f(x)$.
- c** 9 units above the vertex of $f(x)$.
- d** 9 units below the vertex of $f(x)$.